

Backend API

Capability Inventory

Clarification & Baseline — Batch 0

TASK		STATUS			
BE-001		Requirements Baseline / Implementation-Ready After Clarification			
API	REST	BACKEND	Node.js / Express.js		DATABASE PostgreSQL

Purpose

This document establishes the backend requirements and API baseline for the Stock Management System (SMS), based on SRS v2.0. BE-001 inventories backend capabilities, translates them into API surfaces, maps them to FR-01–FR-46 and UC-01–UC-42, and identifies requirements that depend on ASTU policy, applicable Ethiopian government rules, or stakeholder decisions.

BASELINE PRINCIPLE

The backend shall automate approved organizational processes; it shall not invent organizational policy.

Clarification rule: unresolved authority, document model, status, calculation, segregation-of-duty, or relationship questions are BLOCKED / NEEDS CLARIFICATION.

Authority hierarchy & requirement classification

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Authority hierarchy

- 1

Applicable Ethiopian law / government regulation
Use where the operation is governed by a binding public-property, procurement, financial or other applicable rule.
- 2

ASTU approved policy / procedure
Use for ASTU-specific property, receiving, issuing, disposal, inventory and administrative procedures.
- 3

ASTU approved authority / role matrix
Use to determine who may approve, reject, amend, execute or authorize an operation.
- 4

Approved stakeholder decision
Use where the SRS and authoritative policy do not resolve a project-specific question.
- 5

Technical implementation decision
Use only after business requirements are known; developers decide API/database implementation, not policy.

Requirement classification

- BASELINE**

Clearly defined by SRS and safe to implement.
- CONFIGURABLE**

Capability is required, but values/rules should be configurable.
- POLICY-DEPENDENT**

Requires ASTU organizational policy or procedure.
- LEGAL/REGULATORY**

Requires verification against applicable Ethiopian government rules.
- STAKEHOLDER-DECISION**

Requires an explicit project/stakeholder decision.
- BLOCKED**

Cannot safely be finalized until clarification is recorded.
- TECHNICAL**

Implementation can be decided by the development team.

SRS-to-backend domain mapping

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Backend domain → SRS functional requirements

Authentication & IdentityFR-01–FR-04	Stores / DepartmentsFR-05
Categories / Items / UnitsFR-06–FR-07	Suppliers / DonorsFR-08
Locations / BinsFR-09	Receiving / Evaluation / GRNFR-10–FR-16
Stock Cards / Bin CardsFR-17–FR-23	Requisition / Issue / SIV/ISIVFR-24–FR-30
Fixed AssetsFR-31	ReturnsFR-32–FR-34
TransfersFR-35–FR-36	Shelf Life / DisposalFR-37–FR-39
Stock Taking / ControlsFR-40–FR-41	Reports / Notifications / Audit / Dashboard / SearchFR-42–FR-46

IMPLEMENTATION COVERAGE

The mapping above is the backend capability inventory derived from the supplied SRS. The clarification register on the next two pages identifies the business rules that must be confirmed before affected workflows are frozen.

Design implication Implement clear SRS behavior; keep unresolved policy values configurable or mark the affected task BLOCKED.

Clarification register — C-01 to C-09

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ID	Area	SRS mapping	Classification	Required authority / evidence
C-01	Requisition approval authority	FR-25–26 / UC-16	POLICY-DEPENDENT	ASTU authority/role matrix
C-02	TEC approval authority	FR-12–15 / UC-09	POLICY-DEPENDENT	TEC governance
C-03	Which goods require TEC	FR-11–14 / UC-09	POLICY-DEPENDENT	Receiving/property policy
C-04	GRN / Model 19 fields	FR-16 / UC-10	POLICY-DEPENDENT	Official ASTU form/template
C-05	SIV/ISIV document models	FR-27–29 / UC-17–19	POLICY-DEPENDENT	Official forms / policy
C-06	Partial issue rules	FR-30 / UC-17–19	STAKEHOLDER-DECISION	Issue policy
C-07	Amendment of approved/finalized SIV	FR-28 / BR-12	POLICY-DEPENDENT	Document-control policy
C-08	Definition of accepted stock	FR-14–18	POLICY-DEPENDENT	Finance/property policy
C-09	Return disposition	FR-32–34	POLICY-DEPENDENT	ASTU return policy

Clarification register — C-10 to C-17

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ID	Area	SRS mapping	Classification	Required authority / evidence
C-10	Transfer types	FR-35–36	POLICY-DEPENDENT	ASTU transfer policy
C-11	Shelf-life thresholds	FR-37	CONFIGURABLE	Item/category policy
C-12	Disposal authority/evidence	FR-38–39	LEGAL/REGULATORY + POLICY	ASTU + applicable government rules
C-13	FIFO scope	UC-34 / BR-25	POLICY-DEPENDENT	Financial authority confirmation
C-14	Negative-stock exceptions	FR-22	POLICY-DEPENDENT	Stock-control policy
C-15	Document numbering	Multiple	POLICY-DEPENDENT	Document-control policy
C-16	Report layouts	FR-42	POLICY-DEPENDENT	ASTU reporting requirements
C-17	Data migration	NFR-09 / project scope	STAKEHOLDER-DECISION	Approved migration scope

Immediate backend baseline, integrity & API rule

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Immediate Backend Baseline

Authentication

POST /api/v1/auth/login; POST /api/v1/auth/logout; GET /api/v1/auth/me

FR-01-FR-04 / UC-01

Users / RBAC

User CRUD and role/permission administration. Exact authority assignments remain configurable.

FR-02-FR-04 / UC-02

Stores

Store CRUD/status and responsible-officer relationships.

FR-05 / UC-03

Categories / Items

Category and item CRUD, units, status and unique item code.

FR-06-FR-07 / UC-04-UC-05

Locations

Location/bin hierarchy. Exact ASTU physical hierarchy must be confirmed.

FR-09 / UC-07

Receiving

Receipt creation, supporting documents, temporary status and evaluation routing.

FR-10-FR-16 / UC-08-UC-10

Inventory ledger

Stock-affecting operations must pass through validated transactional services: historical balances must not be

FR-17-FR-23 / BR-06-BR-09

TRANSACTION INTEGRITY BASELINE

Stock-affecting operations shall be atomic or recoverable. A transfer must validate source availability, create the outbound and inbound transactions, update derived balances where used, and commit as one logical operation. Failure must roll back or enter a controlled recovery state.

POLICY-DEPENDENT API DESIGN RULE

The API may expose a required capability while keeping exact authorization configurable. Example: POST /api/v1/requisitions/:id/approve may exist because approval is an SRS capability; which role may approve must come from the approved ASTU authority matrix.

Acceptance criteria, sources & final baseline

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BE-001 Acceptance Criteria



- AC-01 — All FR-01–FR-46 have been reviewed and mapped to backend capabilities.
- AC-02 — All UC-01–UC-42 have been reviewed for backend relevance.
- AC-03 — The agreed master backlog contains 150 backend cards.
- AC-04 — Each card identifies its API/capability, FR mapping, UC mapping, actor/permission, dependencies and clarification status.
- AC-05 — Policy-dependent and legal/regulatory requirements are explicitly identified.
- AC-06 — No unresolved organizational rule is silently converted into an authorization rule or irreversible database constraint.
- AC-07 — The clarification register and decision log are prepared for stakeholder validation.

Authority & Source Register

- S1

Stock Management System SRS v2.0

Primary project authority for FR-01–FR-46, UC-01–UC-42, business rules, architecture, acceptance criteria and Batch 0 clarification requirements.
- S2

ASTU Property Administration

ASTU describes property records/inventory, bin cards, stock cards, segregation of duties, fixed assets, disposal, annual inventory and stock inventory.
[Open source](#) ■
- S3

ASTU Procurement and Property Administration context

ASTU states that property administration is carried out according to public procurement and property-administration rules and regulations.
[Open source](#) ■
- S4

Federal Public Procurement and Property Authority (FPPA)

Official property-document repository including Directive No. 1095/2025, Proclamation No. 1333/2024 and stock-management manuals.
[Open source](#) ■

FINAL BASELINE RULE

If the SRS clearly defines the behavior, implement it. If the capability is required but the policy value is unknown, make it configurable or mark it for clarification. If ASTU has an applicable approved policy, follow it. If Ethiopian law or regulation governs the matter, verify the current applicable rule.

Never convert a developer assumption into a permanent business rule.

BASELINE STATUS

Batch 0 requirements and architecture baseline; it does not replace ASTU policy, Ethiopian law, government directives, official templates, financial rules or the final stakeholder authority matrix.